

TNA in Cellular Combustion: Oxidative Agents, Antioxidants, and Metabolic Throughput

Claudio Bresciano – Independent Researcher, Theorem of Axiomatic Necessity (TNA)

Abstract

Cellular metabolism is traditionally modeled via enzymatic pathways and redox balance. This paper applies the **Theorem of Axiomatic Necessity (TNA)**, viewing cellular respiration as an open non-equilibrium system where **throughput (Ψ)** regulates the clearance of **oxidative incoherence (G)**. When G exceeds Ψ , accumulated incoherence ($\aleph$) manifests as oxidative stress, mitochondrial dysfunction, or aging-related pathology. Antioxidants act as external throughput boosters, while metabolic overload saturates Ψ . The model is predictive, falsifiable, and unifies phenomena from caloric restriction to antioxidant interventions.

Keywords: TNA, cellular respiration, oxidative stress, antioxidants, mitochondrial function, throughput, incoherence, metabolic balance

1. Introduction: Cells as Micro-Plumbing Systems

Consider a cell as a combustion engine: glucose and oxygen enter (G), mitochondria process substrates (Ψ), and reactive oxygen species (ROS) are discharged.

$$\frac{d\aleph}{dt} = G(t) - \Psi(t)$$

Where:

- G = oxidative generation (substrate + ROS production)
- Ψ = cellular throughput (mitochondrial efficiency, antioxidant systems, repair)
- $\aleph$ = accumulated oxidative incoherence (damage to DNA, lipids, proteins)

TNA provides a framework to **quantify when oxidative load exceeds cellular capacity**, predicting stress and collapse.

2. Core TNA Mapping to Cellular Combustion

TNA VARIABLE	CELLULAR EQUIVALENT	BIOLOGICAL MANIFESTATION	EXAMPLE CONDITION
G (generation)	Substrate oxidation	ROS, superoxide, hydrogen peroxide	High-fat diet, hyperglycemia
Ψ (throughput)	Antioxidant enzymes, repair, autophagy	SOD, catalase, glutathione, mitophagy	Exercise-induced enhancement, caloric restriction
$\aleph$ (incoherence)	Oxidative damage	DNA lesions, lipid peroxidation, protein carbonylation	Aging, metabolic syndrome
F_min (minimal coherence)	Baseline redox homeostasis	NAD ⁺ /NADH ratio, mitochondrial membrane potential	Viability threshold

3. Applications of TNA in Cellular Oxidation

3.1 Oxidative Stress in Metabolic Overload

- G high (excess calories, toxins) > Ψ low (sedentary, poor antioxidant defense) → $\aleph$ accumulates → inflammation, insulin resistance.

3.2 Antioxidant Interventions

- External Ψ boosters (vitamin C, E, polyphenols) enhance discharge → reduce $\aleph$.
- TNA predicts diminishing returns: excessive antioxidants can saturate feedback and paradoxically reduce endogenous Ψ .

3.3 Caloric Restriction and Mitohormesis

- Mild G restriction → Ψ remains high → controlled ROS act as signaling → $\aleph$ low, longevity increased.
- Empirical data: caloric restriction extends lifespan via **enhanced mitochondrial throughput**.

3.4 Exercise and Cellular Throughput

- Exercise increases Ψ : mitochondrial biogenesis, antioxidant enzyme upregulation.
- Prevents $\aleph$ accumulation despite transient ROS spikes (oxidative hormesis).

3.5 Aging and Mitochondrial Decline

- Ψ declines with age $\rightarrow$ same G generates higher $\aleph \rightarrow$ oxidative damage, senescence.
 - TNA predicts interventions: targeted Ψ boosters, exercise, and caloric modulation reduce cumulative $\aleph$.
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4. Predictive Power and Falsifiability

- Predicts: **throughput-enhancing interventions** outperform simple ROS scavenging for reducing oxidative damage.
 - Falsifiable: If enhancing Ψ fails to reduce $\aleph$ in controlled models, TNA assumptions require revision.
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5. Figure Suggestion: Cellular Oxidative Sacrifice Rate Curve

- **X-axis:** Sacrifice rate (ROS production, metabolic intensity)
 - **Y-axis:** Cellular coherence (redox balance)
 - **Zones:**
 - Left: $\aleph$ accumulation (oxidative stress, apoptosis)
 - Middle: Optimal balance (signaling + repair)
 - Right: Over-catabolism (loss of energy, protein degradation)
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6. Implications for Nutritional and Pharmacological Practice

- Focus on **throughput engineering**: exercise, caloric modulation, intermittent fasting, mitochondrial cofactors.
- External antioxidants act as **throughput boosters**, not primary “fixes.”
- Personalized approach: measure oxidative markers (G/ Ψ ratio) to optimize interventions.

7. Conclusion

- Cellular oxidative stress is not a failure of ROS per se.
 - It is a **throughput failure**: G exceeds Ψ , $\aleph$ accumulates.
 - TNA provides a **unifying mechanical framework** to understand metabolism, oxidative stress, and longevity interventions.
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Canonical Statement

Cells do not fail because they burn energy; they fail when throughput cannot process the byproducts.